

PREPARED FOR

# OUTCAST

SUNNY

T6N, R4W, Mount Diablo Meridian

| COORDINATES                                    | ELEVATION                    | STATE PLANE                                                                              | FEMA ZONE                     | DECLINATION                |
|------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------|-------------------------------|----------------------------|
| <b>38.3459°N</b><br><b>122.3047°W</b><br>WGS84 | <b>66 ft</b><br>NAVD88 · MSL | <b>CA 2 (0402) ·</b><br><b>N 1,887,922.81</b><br><b>E 6,474,289.05</b><br>(US ft, NAD83) | <b>Zone X</b><br>06055C0510FF | <b>13.0° E</b><br>WMM 2026 |

APN 038050009000 | ZONING AP | ACRES 0.481 ac

BEST GNSS WINDOW Tue, Jul 14 · 6:00 AM – 10:00 AM | PDOP 0.98 · 33 sv | NEAREST CORS  
P261 · 14.1 mi ✓

## REPORT CONTENTS

|    |                               |      |                                     |    |
|----|-------------------------------|------|-------------------------------------|----|
| 01 | 7-DAY GNSS FIELD RANKING      | 202  | SITE & SIGNAL REPORT                | 3  |
| 03 | BEST STATIC OBS WINDOW        | 404  | EMERGENCY HOSPITALS                 | 5  |
| 05 | DAY-BY-DAY WEATHER +<br>DRONE | 606  | AI FIELD ANALYSIS                   | 13 |
| 07 | WATER LEVELS & GAUGES         | 1408 | FAULTS & SUBSIDENCE                 | 15 |
| 09 | STRUCTURAL PARAMETERS         | 1610 | SOILS & ENGINEERING                 | 16 |
| 11 | MINES & MINERALS              | 1712 | NATIVE PLANTS & WILDLIFE            | 18 |
| 13 | FCC MICROWAVE PATHS           | 1914 | SOLAR & PASSIVE DESIGN              | 21 |
| 15 | AI SITE INTELLIGENCE<br>BRIEF | 2316 | FIELD MAP · TOPO · DXF<br>SITE PLAN | 24 |



Site file · Google Earth  
Download KMZ

RECORDER  
Napa County Recorder-Clerk · 1127 First  
St, Napa, CA 94559

NEAREST CORS  
P261 · 14.1 mi

GENERATED  
July 14, 2026

ASSESSOR  
Napa County Assessor · 1127 First St,  
Napa, CA 94559

NEAREST BM  
JT9010 · 0.2 mi

SYSTEM  
TopoScout · info@toposcout.com



This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

## SITE & SIGNAL REPORT

### COORDINATES

**38.3459, -122.3047**

WGS84 / NAD83

### ELEVATION

**66 ft**

Above MSL (approx)

### MAGNETIC DECLINATION

**13.0° E**

Compass reads 13.0° too far W

### FEMA FLOOD ZONE

**X**

AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

## IONOSPHERIC CONDITIONS

### PEAK KP INDEX

**4.7**

### AVG KP

**2.4**

### SOLAR FLUX F10.7

**138** sfu

### TEC (VTEC) ▲ DEGRADED

**14.8** TECU

-4.5 TECU vs median

### ASSESSMENT

**MODERATE ACTIVITY**

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

## GNSS / DOP SUMMARY

| DATE        | AVG PDOP | BEST 4-HR WINDOW   | AVG SATS | RATING |
|-------------|----------|--------------------|----------|--------|
| Tue, Jul 14 | 0.98     | 6:00 AM – 10:00 AM | 33       | Ideal  |

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation  
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.



## WILDFIRE PROXIMITY

### No Active Fires Within 150 Miles

No active fires as of report date (July 14, 2026) · Conditions may change – verify before fieldwork



## GPS CONSTELLATION HEALTH

### ▲ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

Land Cover & Multipath Risk

SKY OBSTRUCTION

MULTIPATH RISK

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.9–0.9 across survey window  
PDOP remains ≤ 0.9 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

| Point  | Cover Type  | Obstruction | Multipath |
|--------|-------------|-------------|-----------|
| Center | Dwarf Scrub | Low         | Low       |
| North  | Dwarf Scrub | Low         | Low       |
| South  | Dwarf Scrub | Low         | Low       |
| East   | Dwarf Scrub | Low         | Low       |
| West   | Dwarf Scrub | Low         | Low       |

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

High 99°F | Low 58°F | Wind max 12 mph | Gusts 16 mph | Precip 0.00" | Sunrise 5:57 AM | Sunset 8:32 PM

Waning Crescent 0%

GNSS: Avg PDOP **0.98** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.93, 33 sats avg)

| GNSS / DOP - HOURLY |      |  |       |        |
|---------------------|------|--|-------|--------|
|                     | PDOP |  | SATS  | RATING |
| 6:00 AM             | 0.91 |  | 33 sv | IDEAL  |
| 7:00 AM             | 1.00 |  | 32 sv | GOOD   |
| 8:00 AM             | 0.88 |  | 35 sv | IDEAL  |
| 9:00 AM             | 0.93 |  | 33 sv | IDEAL  |
| 10:00 AM            | 1.09 |  | 30 sv | GOOD   |
| 11:00 AM            | 1.03 |  | 29 sv | GOOD   |
| 12:00 PM            | 0.92 |  | 31 sv | IDEAL  |
| 1:00 PM             | 1.20 |  | 24 sv | GOOD   |
| 2:00 PM             | 0.97 |  | 32 sv | IDEAL  |
| 3:00 PM             | 0.93 |  | 33 sv | IDEAL  |
| 4:00 PM             | 0.97 |  | 33 sv | IDEAL  |
| 5:00 PM             | 0.98 |  | 34 sv | IDEAL  |
| 6:00 PM             | 1.03 |  | 28 sv | GOOD   |
| 7:00 PM             | 0.92 |  | 31 sv | IDEAL  |



GO - GOOD CONDITIONS

Partly cloudy | Wind: 12.3 mph (gusts 16.1 mph) | Precip: 0" | Temp: 57.8°–98.7°F



FIELD SAFETY WARNINGS



HIGH HEAT 98.7°F — Monitor crew for heat stress



DRONE WIND ANALYSIS



GO Peak: 12.3/16.1 mph S

\* Sunrise: 5:57 AM | Sunset: 8:32 PM | fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

|       | SUSTAINED / GUSTS mph | ≤ turbulence | DIR |
|-------|-----------------------|--------------|-----|
| 5 AM  | 5/6 mph               |              | SE  |
| 6 AM  | 5/7 mph               |              | ESE |
| 7 AM  | 4/5 mph               |              | ESE |
| 8 AM  | 1/1 mph               |              | S   |
| 9 AM  | 3/3 mph               |              | S   |
| 10 AM | 4/4 mph               |              | S   |
| 11 AM | 6/7 mph               |              | S   |
| 12 PM | 7/7 mph               |              | S   |
| 1 PM  | 8/9 mph               |              | S   |
| 2 PM  | 10/12 mph             |              | SSW |
| 3 PM  | 12/16 mph             |              | SSW |
| 4 PM  | 12/16 mph             |              | SSW |
| 5 PM  | 9/13 mph              |              | WSW |
| 6 PM  | 6/11 mph              |              | WSW |
| 7 PM  | 2/13 mph              |              | SW  |
| 8 PM  | 9/12 mph              |              | SE  |

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

**FIELD PLANNING REPORT**

JOB: SG-GSVJSH | SITE: N/A | LOCATION: 38.3459, -122.3047 (Napa County, CA)

REPORT DATE: Tuesday, July 14, 2026 | SURVEY START: Tuesday, July 14, 2026

DAY RANKING & VERDICT

VERDICT: GO

REASON: Excellent PDOP geometry (0.93–0.98 range), clear weather with low wind, and moderate ionospheric conditions manageable via extended occupation. No weather barriers present.

**BEST SURVEY WINDOW**

TIME WINDOW: 0600–1000 UTC (Tuesday, 07 Jul 2026)

- PDOP: 0.93 (best of day)
- Satellites: 33 multi-constellation (GPS/GLONASS/Galileo/BeiDou)
- Duration: 4 hours
- Rationale: Peak geometry window. Extended static occupation recommended due to moderate Kp activity (peak 4.7) and below-median TEC (–4.5 TECU). Four-hour session will mitigate ionospheric delay.

**BEST 4-HOUR STATIC GNSS WINDOW**

WINDOW: 0600–1000 UTC

- PDOP: 0.93
- Satellite count: 33
- Expected accuracy:  $\pm 0.05$ – $0.08$  m horizontal (post-processed, multi-base RTK or static with ionosphere-free combinations)
- Ionospheric mitigation: Moderate activity (Kp 4.7 avg) — use multi-frequency receivers and ionosphere-free linear combinations; four-hour occupation adequate for sub-decimeter static positioning

**DRONE OPERATIONS ASSESSMENT**

CLEARED FOR VFR OPERATIONS. Wind max 12 mph is well within limits for sUAS (Part 107 threshold 19 mph). Clear skies, 10+ km visibility, no precipitation. Recommend flight ops during 0800–1400 local time window for optimal lighting and stable thermal conditions.

**FIELD SAFETY WARNINGS**

- SATELLITE HEALTH: GPS PRN 11 currently unhealthy (UNUSUFN) — receivers will auto-exclude. No impact on PDOP forecast.
- IONOSPHERIC CONDITIONS: Moderate geomagnetic activity present. If static baseline > 20 km, strongly recommend dual-frequency receivers and extended occupation ( $\geq 4$  hours) or use network corrections.
- HEAT: Forecast high 99°F. Ensure adequate hydration and sun protection during midday operations (1000–1600 local).

**SITE CONDITIONS SUMMARY**

Napa County location shows low multipath risk (dwarf scrub land cover, low sky obstruction). Recommended elevation mask 13° based on terrain. No active wildfires detected within range. Air quality good (AQI 44). These conditions support clean GNSS observations.

**EXECUTIVE SUMMARY**

Tuesday, July 14, 2026 is a GO day for GNSS survey. Geometry is excellent (PDOP 0.93–0.98), weather is clear with light winds (max 12 mph, 99°F), and ionospheric activity is moderate but manageable. Execute primary static observations during the 0600–1000 UTC window using multi-frequency receivers and four-hour minimum occupation to account for Kp 4.7 activity and below-median TEC. Drone operations are VFR-approved. Monitor for heat stress and note GPS PRN 11 unhealthy status (auto-excluded by receivers). Site multipath environment is favorable.

**COUNTY RECORDING OFFICES**

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk

1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor

1127 First St, Napa, CA 94559

CORS STATIONS – WITHIN 30 MILES

| QR / LINK                                                                                  | STATION ID | NAME / LOCATION  | DISTANCE |
|--------------------------------------------------------------------------------------------|------------|------------------|----------|
| <br>NGS ↗ | P261       | HUNTERHILLCN2004 | 14.1 mi  |
| <br>NGS ↗ | P198       | PETALUMAIRC2004  | 17.5 mi  |
| <br>NGS ↗ | OHLN       | OHLONE PARK      | 23.5 mi  |
| <br>NGS ↗ | P196       | MEACHUMLFLCN2006 | 24.0 mi  |
| <br>NGS ↗ | CASR       | SANTA ROSA CA    | 24.9 mi  |
| <br>NGS ↗ | P267       | DIXONAVIATCN2005 | 26.3 mi  |

NGS BENCHMARKS – WITHIN 10 MILES

| QR / LINK                                                                                          | PID    | DESIGNATION | ELEVATION               | ORDER | DISTANCE |
|----------------------------------------------------------------------------------------------------|--------|-------------|-------------------------|-------|----------|
| <br>Datasheet ↗  | JT9010 | "13"        | 19.100m / 62.7ft NAVD88 | Order | 0.2 mi   |
| <br>Datasheet ↗ | JT9018 | "13 E"      | 17.970m / 59.0ft NAVD88 | Order | 0.2 mi   |
| <br>Datasheet ↗ | JT9009 | "13"        | 17.900m / 58.7ft NAVD88 | Order | 0.2 mi   |
| <br>Datasheet ↗ | JT9017 | "13 D"      | 17.890m / 58.7ft NAVD88 | Order | 0.2 mi   |
| <br>Datasheet ↗ | JT9034 | "20 A"      | 20.730m / 68.0ft NAVD88 | Order | 0.3 mi   |
| <br>Datasheet ↗ | JT9012 | "13"        | 19.060m / 62.5ft NAVD88 | Order | 0.4 mi   |
| <br>Datasheet ↗ | JT9043 | "25 A"      | 18.710m / 61.4ft NAVD88 | Order | 0.4 mi   |
| <br>Datasheet ↗ | JT9016 | "13 D"      | 18.200m / 59.7ft NAVD88 | Order | 0.5 mi   |

| QR /<br>LINK   | PID | DESIGNATION | ELEVATION | ORDER | DISTANCE |
|----------------|-----|-------------|-----------|-------|----------|
| Datasheet<br>↗ |     |             |           |       |          |

FEMA FLOOD ZONE

**X – AREA OF MINIMAL FLOOD HAZARD**

Panel: 06055C0510FF

Effective: Sep 29, 2010

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

| PID    | NAME | LAT       | LON         | ELEV_FT | ORDER |
|--------|------|-----------|-------------|---------|-------|
| JT9010 | 13   | 38.348333 | -122.302778 | 62.7    |       |
| JT9018 | 13 E | 38.348333 | -122.302778 | 59.0    |       |
| JT9009 | 13   | 38.346667 | -122.301111 | 58.7    |       |
| JT9017 | 13 D | 38.345000 | -122.301111 | 58.7    |       |
| JT9034 | 20 A | 38.345000 | -122.309444 | 68.0    |       |
| JT9012 | 13   | 38.351667 | -122.304444 | 62.5    |       |
| JT9043 | 25 A | 38.340000 | -122.302778 | 61.4    |       |
| JT9016 | 13 D | 38.341667 | -122.297778 | 59.7    |       |

Download .txt File



## USGS STREAM GAUGES – within 25 miles

| STATION                               | DIST    | LEVEL    | FLOW     | FLOOD STATUS |
|---------------------------------------|---------|----------|----------|--------------|
| NAPA R NR NAPA CA #11458000           | 1.5 mi  | 2.36 ft  | 4.08 cfs | ● Normal     |
| NATHANSON C A SONOMA CA #11458600     | 9.3 mi  | —        | —        | ● Normal     |
| SONOMA C A AGUA CALIENTE CA #11458500 | 10.4 mi | 8.83 ft  | 2.6 cfs  | ● Normal     |
| NAPA R NR ST HELENA CA #11456000      | 14.1 mi | 1.43 ft  | 3.23 cfs | ● Normal     |
| SONOMA CREEK A KENWOOD CA #11458433   | 14.9 mi | 3.92 ft  | 0.11 cfs | ● Normal     |
| PUTAH C NR WINTERS CA #11454000       | 16.8 mi | 8.03 ft  | 641 cfs  | — Unknown    |
| SUISUN SLOUGH NR BENICIA CA #11455740 | 19.6 mi | 13.47 ft | -588 cfs | — Unknown    |
| POPE C A WALTER SPRINGS CA #11453590  | 21.1 mi | —        | —        | — Unknown    |

## USGS LAKES &amp; RESERVOIRS – within 150 miles

| STATION                          | DIST    | LEVEL   | FULL POOL | LINK                   |
|----------------------------------|---------|---------|-----------|------------------------|
| CLEAR LK A LAKEPORT CA #11450000 | 58.1 mi | 4.99 ft | N/R       | <a href="#">USGS ↗</a> |

## CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

| RESERVOIR        | DIST     | ELEVATION            | FULL POOL | STORAGE   | % CAP |
|------------------|----------|----------------------|-----------|-----------|-------|
| Lake Del Valle   | 60.6 mi  | 696.82 ft (-9 ft)    | 706 ft    | 36K af    | 67%   |
| Folsom Lake      | 66.5 mi  | 447.42 ft (-19 ft)   | 466 ft    | 780K af   | 80%   |
| Lake Oroville    | 93.4 mi  | 859.55 ft (-40 ft)   | 900 ft    | 2,836K af | 80%   |
| New Melones Lake | 100.1 mi | 1,019.82 ft (-68 ft) | 1,088 ft  | 1,660K af | 69%   |
| Black Butte Lake | 101.2 mi | —                    | 228 ft    | 73K af    | 54%   |

## USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

| RESERVOIR | DIST | ELEVATION | FULL POOL | STORAGE | % CAP |
|-----------|------|-----------|-----------|---------|-------|
|-----------|------|-----------|-----------|---------|-------|

## ARMY CORPS RESERVOIRS (USACE) – within 150 miles

| RESERVOIR             | DIST   | POOL ELEV | FULL POOL | STORAGE | % CAP |
|-----------------------|--------|-----------|-----------|---------|-------|
| Del Valle             | 64 mi  | 696.8 ft  | 695.01 ft | —       | 100%  |
| Camanche              | 89 mi  | 230.2 ft  | 221.86 ft | —       | 104%  |
| Englebright Lake-Pool | 95 mi  | 525.1 ft  | 522.35 ft | —       | 101%  |
| Farmington Dam-Pool   | 99 mi  | 125.3 ft  | 117.09 ft | —       | 107%  |
| Black Butte-Pool      | 101 mi | 456.8 ft  | 458.36 ft | —       | 100%  |
| Don Pedro             | 137 mi | —         | N/R       | —       | —     |

## USGS GROUNDWATER WELLS – within 15 miles

| STATION         | DIST   | WELL DEPTH | WATER LEVEL  | DATE       | AQUIFER    |
|-----------------|--------|------------|--------------|------------|------------|
| 005N004W32A002M | 7 mi   | 200 ft     | 87.71 ft BGS | 2014-11-05 | N100CACSTL |
| 005N004W28R001M | 7 mi   | 51 ft      | 41.90 ft BGS | 1977-10-03 | N100CACSTL |
| 004N004W04C001M | 8 mi   | 80 ft      | 11.20 ft BGS | 1977-10-03 | N100CACSTL |
| 004N004W05B001M | 8 mi   | —          | 54.10 ft BGS | 1977-10-03 | N100CACSTL |
| 005N004W31R001M | 8.1 mi | 295 ft     | —            | —          | —          |
| 004N004W05D002M | 8.3 mi | 60 ft      | 14.60 ft BGS | 1977-10-03 | N100CACSTL |

|                 |        |        |              |            |            |
|-----------------|--------|--------|--------------|------------|------------|
| 004N004W06B001M | 8.3 mi | 201 ft | 21.86 ft BGS | 2022-06-07 | —          |
| 004N004W02L001M | 8.8 mi | —      | 9.90 ft BGS  | 1977-10-02 | N100CACSTL |

🕒 AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Napa Valley Coastal Ranges aquifer (N100CACSTL) consists of Tertiary-age sandstone, siltstone, and shale deposited in marine and fluvial environments during the Miocene to Pliocene epochs. This formation underlies much of Napa County and yields water from fractured and intergranular porosity at moderate depths. Water availability at this site is moderate, with the water table ranging from roughly 10 to 88 feet below ground surface depending on seasonal recharge and pumping demand.

FORMATION

|            |                                                                          |
|------------|--------------------------------------------------------------------------|
| Name       | Napa Valley Coastal Ranges Tertiary Sandstone and Siltstone (N100CACSTL) |
| Age / Rock | Miocene to Pliocene sandstone, siltstone, and shale                      |
| Extent     | 800–1200 sq mi (est. USGS/DWR)                                           |

DEPTH & RECHARGE

|                 |                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------|
| Depth Range     | 9.90–295 ft (1–30 floors)                                                                                                     |
| Typical Depth   | 50–150 ft (5–15 floors)                                                                                                       |
| Recharge Source | Precipitation on outcrops and hillslopes of the Mayacamas and Diablo Ranges; minor lateral inflow from surrounding formations |
| Transit Time    | 15–40 years to 100 ft depth (est. USGS/DWR)                                                                                   |

WATER QUALITY

|          |                                                                                                                                                                         |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pH       | est. 6.8–7.4 (est. USGS/DWR)                                                                                                                                            |
| Hardness | est. 180–350 mg/L CaCO3 (est. USGS/DWR)                                                                                                                                 |
| TDS      | est. 250–500 mg/L (est. USGS/DWR)                                                                                                                                       |
| Notes    | No sample data available — typical for formation. Regional groundwater suitable for domestic and agricultural use; minor iron and manganese possible in reducing zones. |

TREND

|                |                                                                                                                                                                                                                                         |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trend          | Water levels show long-term decline and increased seasonal variability. 1977 measurements (9.9–54.1 ft) versus 2022 (21.86 ft) and 2014 (87.71 ft) indicate multi-decadal drawdown and possible aquifer stress from cumulative pumping. |
| Source         | USGS NWIS well observations on file                                                                                                                                                                                                     |
| Seasonal Swing | est. 10–30 ft annual swing (est. USGS/DWR); exact range at this site subject to local recharge and pumping patterns                                                                                                                     |

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The Castle Rock area (N100CACSTL aquifer) is located outside any critically overdrafted groundwater basin and exhibits well-drained soils typical of consolidated volcanic and metamorphic terrain in the northern California Coast Ranges. The absence of extensive clay aquitards and the site's position outside SGMA-designated basins indicate minimal historical groundwater withdrawal stress and negligible aquifer compaction. This means surveyors working in this area can rely on stable vertical datum references without subsidence adjustments to their elevations.

SUBSIDENCE ASSESSMENT

|                    |                                    |
|--------------------|------------------------------------|
| Susceptibility     | Low                                |
| Primary Mechanism  | None - consolidated terrain        |
| Annual Rate        | <0.5 mm/yr (est. USGS)             |
| Historic Loss      | None documented                    |
| Aquifer Compaction | None - aquifer not over-pumped     |
| SGMA Status        | Not applicable - not in SGMA basin |

BENCHMARK IMPLICATIONS

|                     |                                          |
|---------------------|------------------------------------------|
| Monument Risk       | Low - monuments stable                   |
| Elevation Validity  | Good - no active subsidence              |
| GNSS Recommendation | Standard benchmark verification adequate |

SURVEY ACTION

|                 |                                                                          |
|-----------------|--------------------------------------------------------------------------|
| Required Action | No special action required                                               |
| Dataset         | CA DWR SGMA Bulletin 118 2024.1                                          |
| Coverage        | California critically overdrafted basins only - full US coverage pending |

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

|               |             |
|---------------|-------------|
| Site Class    | D (default) |
| Risk Category | II          |

WIND / SNOW / ICE / FROST (ASCE 7-22)

|                                                         |               |
|---------------------------------------------------------|---------------|
| Basic Wind Speed                                        | 92 mph        |
| Ground Snow Load                                        | 5.8 psf       |
| Winter Wind (W2)                                        | 0.35          |
| Ice Thickness                                           | None          |
| Frost Depth                                             | 12 in minimum |
| Est. from IECC Zone 3C — AFI data unavailable           |               |
| Verify with local building official before construction |               |

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This marine climate near the California coast stays mild year-round with moderate cooling needs and minimal heating, so your envelope can prioritize air sealing and moisture management over heavy insulation—the real challenge is preventing condensation and salt air degradation rather than temperature extremes. Maximize south and west-facing glazing for passive solar gains in winter, but shade it carefully or use high-performance glass to prevent summer overheating, while keeping north-facing glass modest to reduce heat loss and air infiltration on the ocean-facing side. Your key passive move is designing deep overhangs and operable windows to capture cool ocean breezes for natural ventilation in summer, which will be far more effective than mechanical cooling in this climate.

AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This coastal marine site in IECC Zone 3C presents well-drained gravelly soils suitable for conventional shallow foundations. Seismic parameters are currently unavailable for this location; a geotechnical investigation and seismic hazard screening are recommended early in design to establish appropriate structural detailing. Wind exposure and moderate snow loads suggest standard lateral and roof design approaches, though coastal conditions may warrant verification of exposure classification.

FOUNDATION

|       |                                                                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type  | Conventional spread footing anticipated                                                                                                                                                                                                                                                                  |
| Basis | Well-drained gravelly loam (Coombs series) and gentle slopes (2–5%) suggest adequate bearing capacity for shallow foundations. Consult a licensed geotechnical engineer to confirm allowable bearing pressure and settlement estimates; seismic site class confirmation may influence foundation design. |

STRUCTURAL IMPLICATIONS

|               |                                                                                                                                                                                                                                                                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seismic       | Seismic design category and spectral parameters are unavailable for this location. A seismic hazard assessment and site-specific ground motion analysis are suggested to determine if ordinary or special lateral-force-resisting systems are appropriate. Site Class D (planned assumption) may warrant soil profile confirmation via boring. |
| Wind Exposure | Exposure C (open terrain with scattered obstructions) anticipated per ASCE 7-22 Risk Category II. Coastal location may warrant review of terrain classification and exposure category; consult engineer if site topography or nearby wind breaks differ materially.                                                                            |
| Wind          | 92 mph 3-second gust speed (Risk Cat II, Exposure C) suggests standard wind load design. No hurricane or extreme coastal wind zone indicated; however, proximity to marine influence may warrant confirmation of exposure assumptions.                                                                                                         |
| Snow          | Ground snow load of 5.8 psf is light; standard roof design and typical snow load provisions likely adequate. Winter wind factor W2 = 0.35 suggests minimal snow drifting—consult engineer if roof configuration creates drift pockets.                                                                                                         |

INVESTIGATIONS & INSPECTIONS

|                     |                                                                                                                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Geotech Required    | Yes—geotechnical investigation recommended to confirm Site Class D assumption, establish bearing capacity and settlement parameters, and screen for liquefaction potential given coastal setting and well-drained soils.                               |
| Liquefaction Screen | Screening recommended despite low subsidence risk. Well-drained gravelly soils (Hydro Group C) suggest low liquefaction susceptibility, but shallow groundwater and coastal proximity may warrant a Phase 1 environmental and geotechnical assessment. |
| Special Inspections | Not anticipated based on available parameters; however, special inspections may be required once seismic design category is established. Consult engineer and building official after seismic hazard assessment.                                       |
| Code Edition        | ASCE 7-22 / IBC 2021 (planning reference only)                                                                                                                                                                                                         |
| Site Class Note     | Site Class D assumed for planning based on soil description. Geotechnical investigation with standard penetration testing and shear wave velocity measurement recommended to confirm site class per ASCE 7-22 Section 20.4.                            |

SOILS — USDA SOIL SURVEY

|                                                       |                   |
|-------------------------------------------------------|-------------------|
| MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes | SERIES: Coombs    |
| DRAINAGE CLASS: Well drained                          | HYDRO GROUP: C    |
| SOIL ORDER: Alfisols                                  | SUBORDER: Xeralfs |

AI SOIL INTELLIGENCE — CLAUDE HAIKU

Coombs gravelly loam is a well-drained Alfisol (Xeralf) formed in mixed alluvial and colluvial materials on gentle hillslopes in Mediterranean climates. The soil profile features a gravelly loam surface layer overlying a clay-enriched substratum, typical of soils developed under seasonal moisture stress. Surveyors and developers should expect moderate gravel content, moderate shrink-swell potential in clay-rich subsoils, and generally stable conditions for light to moderate foundation work on these 2–5% slopes.

Mixed alluvial and colluvial parent material; foothill and lower bajada positions; oak and chaparral vegetation zones.

DRAINAGE

|                     |                        |
|---------------------|------------------------|
| Class               | Well drained           |
| Seasonal Saturation | None within 60 inches  |
| Flooding Risk       | Minimal on 2–5% slopes |
| Ponding Risk        | None                   |

ENGINEERING

|                  |                                                                   |
|------------------|-------------------------------------------------------------------|
| Bearing Capacity | 2,500–3,500 psf (shallow foundations)                             |
| Rating           | Fair                                                              |
| Shrink-Swell     | Moderate — clay substratum (typical of Alfisols)                  |
| Frost Action     | Low                                                               |
| Cut Suitability  | Good (gravelly material excavates readily)                        |
| Fill Suitability | Fair (clay substratum — moisture control and compaction critical) |

SEPTIC

|             |                                                                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Suitability | Conditional                                                                                                                                     |
| Notes       | Perc test required — clay loam substratum (argillic horizon) may limit permeability; seasonal water table and slope gradient should be verified |

HAZARDS

|            |                                                              |
|------------|--------------------------------------------------------------|
| Excavation | Moderate — 15–30% gravel content may slow equipment progress |
|------------|--------------------------------------------------------------|

|                |                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------|
| Expansive Clay | Moderate — argillic horizon exhibits clay accumulation; monitor for volume change during fill operations                |
| Erosion        | Low on gentle slopes; moderate risk in disturbed or denuded areas; gully formation possible if drainage is concentrated |
| Other          | None anticipated                                                                                                        |

Data sources: USGS National Water Information System · [waterdata.usgs.gov](#) · California DWR / CDEC · [cdec.water.ca.gov](#) · USBR RISE · [data.usbr.gov](#) · USACE CWMS · [water.usace.army.mil](#) · USGS NWIS Groundwater · [waterservices.usgs.gov](#) · USDA Web Soil Survey · [sdmdataaccess.nrcs.usda.gov](#) · CA DWR SGMA Bulletin 118 · [gis.water.ca.gov](#) · ASCE 7-22 Hazard Tool · [gis.asce.org](#) · USGS Design Maps · [earthquake.usgs.gov](#) · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision

Water levels current as of report run: Jul 14, 2026, 5:14 AM UTC

\* Live pool elevation not reported by this reservoir — full pool design elevation shown

 **OIL & GAS WELLS** California — within 10mi radius

8

total

0

active

7

plugged

By status:

7

Plugged

1

Idle

By type: 8 DH

| Lease / Well              | Status  | Type | Operator                  | Depth | Dist   |
|---------------------------|---------|------|---------------------------|-------|--------|
| Mobil Oil Corporation     | Plugged | DH   | Mobil Oil Corporation     | —     | 7.2 mi |
| Mobil Oil Corporation     | Plugged | DH   | Mobil Oil Corporation     | —     | 7.2 mi |
| McKenzie-Hazard Syndicate | Plugged | DH   | McKenzie-Hazard Syndicate | —     | 7.4 mi |
| Mobil Oil Corporation     | Plugged | DH   | Mobil Oil Corporation     | —     | 8.6 mi |
| Mobil Oil Corporation     | Plugged | DH   | Mobil Oil Corporation     | —     | 8.8 mi |
| Mobil Oil Corporation     | Plugged | DH   | Mobil Oil Corporation     | —     | 9.5 mi |
| Capell Oil Corp.          | Idle    | DH   | Capell Oil Corp.          | —     | 9.7 mi |
| Cordelia Oil & Gas Co.    | Plugged | DH   | Cordelia Oil & Gas Co.    | —     | 12 mi  |

**AI Assessment — Well Proximity**

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil Corporation dry hole located 7.2 miles away, well beyond any actionable distance threshold. No active producers, injection wells, or other well types exist within relevant distances that would require foundation review, environmental assessment, or surface rights investigation.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

**200**                      **10**                      **108**  
total in search area      within 5 mi              within 25 mi

| Site Name                         | Status        | Commodities   | Distance |
|-----------------------------------|---------------|---------------|----------|
| Roderick Sand and Gravel Pit      | Occurrence    | Sand/Gravel   | 1.6 mi   |
| Basalt Rock Co                    | Producer      | Clay          | 2 mi     |
| Cicero Pumice Quarry              | Occurrence    | PUM           | 2.9 mi   |
| Mount George King Pumice Deposit. | Occurrence    | PUM           | 2.9 mi   |
| Basalt Rock Co.                   | Occurrence    | DIT           | 3.1 mi   |
| Mt George Pumice Deposit          | Producer      | PUM           | 3.1 mi   |
| Wilson Pumice Quarry              | Occurrence    | PUM           | 3.7 mi   |
| Unknown                           | Occurrence    | Mercury       | 4.4 mi   |
| Silva Pumice Quarry               | Occurrence    | PUM           | 4.7 mi   |
| Silva Pumice Quarry               | Producer      | PUM           | 5 mi     |
| Walker Pumice Quarry              | Occurrence    | PUM           | 5.1 mi   |
| Walker Pumice Quarry              | Producer      | PUM           | 5.1 mi   |
| Mountain                          | Occurrence    | Mercury       | 5.4 mi   |
| Unknown                           | Occurrence    | Mercury       | 7.6 mi   |
| Unknown                           | Occurrence    | Stone/Crushed | 7.8 mi   |
| Unnamed Location                  | Past Producer | Sand/Gravel   | 8.5 mi   |
| Unnamed Location                  | Past Producer | Sand/Gravel   | 8.5 mi   |
| Unnamed Prospect                  | Occurrence    | Chromium      | 8.8 mi   |
| Unnamed Location                  | Occurrence    | Chromium      | 9 mi     |
| Unknown                           | Occurrence    | Chromium      | 9.1 mi   |
| Bella Oak                         | Past Producer | Mercury       | 9.4 mi   |
| Unknown                           | Occurrence    | MG            | 9.7 mi   |
| Serres Pit                        | Past Producer | Sand/Gravel   | 10.4 mi  |
| Unnamed Location                  | Unknown       | Sand/Gravel   | 10.5 mi  |
| Conn Valley                       | Occurrence    | Manganese     | 10.6 mi  |

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

#### AI Assessment — Mining Context

Three active mineral producers operate within 5 miles of the site: Basalt Rock Co (2 mi, clay extraction), Mt George Pumice Deposit (3.1 mi, pumice), and Silva Pumice Quarry (5 mi, pumice). Expect heavy truck traffic on access roads during operating hours from all three operations, plus dust generation from the aggregate/clay operations and potential noise from crushing or processing activities at the pumice facilities. No inactive deposits warrant Phase 1 assessment consideration based on proximity.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

#### ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

**6**                      **6**                      **12**  
Endangered      Threatened      Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

| COMMON NAME                        | SCIENTIFIC NAME                                    | STATUS     |
|------------------------------------|----------------------------------------------------|------------|
| California tiger Salamander        | <i>Ambystoma californiense</i>                     | Endangered |
| Conservancy fairy shrimp           | <i>Branchinecta conservatio</i>                    | Endangered |
| Contra Costa goldfields            | <i>Lasthenia conjugens</i>                         | Endangered |
| Soft bird's-beak                   | <i>Cordylanthus mollis</i> ssp. <i>mollis</i>      | Endangered |
| Suisun thistle                     | <i>Cirsium hydrophilum</i> var. <i>hydrophilum</i> | Endangered |
| Vernal pool tadpole shrimp         | <i>Lepidurus packardii</i>                         | Endangered |
| Alameda whipsnake (=striped racer) | <i>Masticophis lateralis euryxanthus</i>           | Threatened |
| California red-legged frog         | <i>Rana draytonii</i>                              | Threatened |
| Delta smelt                        | <i>Hypomesus transpacificus</i>                    | Threatened |
| Northern spotted owl               | <i>Strix occidentalis caurina</i>                  | Threatened |
| Vernal pool fairy shrimp           | <i>Branchinecta lynchi</i>                         | Threatened |
| western snowy plover               | <i>Charadrius nivosus nivosus</i>                  | Threatened |

**AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU**

ESA Implications Review – 38.3459, -122.3047 **ESA Section 7 Triggers & Planning-Relevant Species:** This site's location within critical habitat for 12 species—particularly the endangered California tiger salamander, Conservancy fairy shrimp, and Contra Costa goldfields, along with threatened California red-legged frog and vernal pool fairy shrimp—will trigger mandatory Section 7 consultation if your project involves federal permitting, funding, or oversight. The vernal pool complex and associated seasonal wetland habitats are the most planning-sensitive features, as multiple listed species depend on these ephemeral systems; ground disturbance, drainage modifications, or grading pose the greatest conflict risk. Development footprints, stormwater management, and habitat fragmentation should be the primary design constraints during site planning. **Recommended Next Step:** Retain a qualified environmental biologist to conduct a Phase 1 habitat assessment and presence/absence surveys for vernal pool species during appropriate seasonal windows (typically winter-spring); concurrently, coordinate with USFWS and CDFW to determine whether your anticipated project activities constitute "take" and what incidental take permits or habitat mitigation may be required. ---

\*For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.\*

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

**NATIVE PLANTS - iNaturalist Research-Grade within 10mi**

|                |                    |                   |
|----------------|--------------------|-------------------|
| <b>58</b>      | <b>5052</b>        | <b>2026-07-04</b> |
| Unique Species | Total Observations | Most Recent       |

| COMMON NAME                | SCIENTIFIC NAME                         | LAST OBS   |
|----------------------------|-----------------------------------------|------------|
| American black nightshade  | <i>Solanum americanum</i>               | 2026-05-29 |
| Bearded Clover             | <i>Trifolium barbigerum</i>             | 2026-05-15 |
| bird's foot cliffbrake     | <i>Pellaea mucronata</i>                | 2026-04-28 |
| Blow Wives                 | <i>Achyrachaena mollis</i>              | 2026-05-20 |
| blue oak                   | <i>Quercus douglasii</i>                | 2026-05-09 |
| Bolander's Phacelia        | <i>Phacelia bolanderi</i>               | 2026-04-27 |
| box elder                  | <i>Acer negundo</i>                     | 2026-05-18 |
| Braun's Giant Horsetail    | <i>Equisetum telmateia braunii</i>      | 2026-06-22 |
| California bay             | <i>Umbellularia californica</i>         | 2026-06-24 |
| California beepplant       | <i>Scrophularia californica</i>         | 2026-05-20 |
| California black oak       | <i>Quercus kelloggii</i>                | 2026-05-15 |
| California buckeye         | <i>Aesculus californica</i>             | 2026-06-02 |
| California Dutchman's Pipe | <i>Aristolochia californica</i>         | 2026-05-26 |
| California milkwort        | <i>Rhinotropis californica</i>          | 2026-05-15 |
| California mugwort         | <i>Artemisia douglasiana</i>            | 2026-06-02 |
| California poppy           | <i>Eschscholzia californica</i>         | 2026-06-24 |
| California Wild Rose       | <i>Rosa californica</i>                 | 2026-05-09 |
| Canyon liveforever         | <i>Dudleya cymosa</i>                   | 2026-05-30 |
| -                          | <i>Castilleja densiflora densiflora</i> | 2026-05-09 |
| clay mariposa lily         | <i>Calochortus argillosus</i>           | 2026-05-15 |
| coast live oak             | <i>Quercus agrifolia</i>                | 2026-05-13 |
| coastal woodfern           | <i>Dryopteris arguta</i>                | 2026-05-20 |
| common selfheal            | <i>Prunella vulgaris</i>                | 2026-06-22 |
| common yarrow              | <i>Achillea millefolium</i>             | 2026-05-18 |
| Diogenes' lantern          | <i>Calochortus amabilis</i>             | 2026-05-15 |
| distant phacelia           | <i>Phacelia distans</i>                 | 2026-05-15 |
| -                          | <i>Downingia concolor concolor</i>      | 2026-05-15 |
| Dwarf Brodiaea             | <i>Brodiaea terrestris</i>              | 2026-05-15 |

| COMMON NAME                      | SCIENTIFIC NAME                   | LAST OBS   |
|----------------------------------|-----------------------------------|------------|
| Elegant Clarkia                  | <i>Clarkia unguiculata</i>        | 2026-05-16 |
| goldback fern                    | <i>Pentagramma triangularis</i>   | 2026-05-20 |
| Ithuriel's Spear                 | <i>Triteleia laxa</i>             | 2026-06-15 |
| maroonspot calicoflower          | <i>Downingia concolor</i>         | 2026-05-15 |
| narrowleaf milkweed              | <i>Asclepias fascicularis</i>     | 2026-06-25 |
| narrowleaf mule-ears             | <i>Wyethia angustifolia</i>       | 2026-05-15 |
| Narrowleaf Willow                | <i>Salix exigua</i>               | 2026-06-24 |
| needleleaf navarretia            | <i>Navarretia intertexta</i>      | 2026-05-15 |
| northern California black walnut | <i>Juglans hindsii</i>            | 2026-06-24 |
| orange bush monkeyflower         | <i>Diplacus aurantiacus</i>       | 2026-05-29 |
| Oregon Ash                       | <i>Fraxinus latifolia</i>         | 2026-06-02 |
| Pacific Bleeding Heart           | <i>Dicentra formosa</i>           | 2026-05-04 |
| Pacific madrone                  | <i>Arbutus menziesii</i>          | 2026-07-04 |
| Pacific poison oak               | <i>Toxicodendron diversilobum</i> | 2026-06-02 |
| pineapple-weed                   | <i>Matricaria discoidea</i>       | 2026-05-20 |
| Redwood Sorrel                   | <i>Oxalis smalliana</i>           | 2026-05-28 |
| small-headed clover              | <i>Trifolium microcephalum</i>    | 2026-05-15 |
| Superb Mariposa Lily             | <i>Calochortus superbus</i>       | 2026-06-11 |
| thimble clover                   | <i>Trifolium microdon</i>         | 2026-05-15 |
| tomcat clover                    | <i>Trifolium willdenovii</i>      | 2026-05-15 |
| Toyon                            | <i>Heteromeles arbutifolia</i>    | 2026-06-19 |
| trailing blackberry              | <i>Rubus ursinus</i>              | 2026-06-24 |
| Tree Clover                      | <i>Trifolium ciliolatum</i>       | 2026-05-15 |
| valley oak                       | <i>Quercus lobata</i>             | 2026-06-24 |
| wavy-leafed soap plant           | <i>Chlorogalum pomeridianum</i>   | 2026-05-20 |
| western sycamore                 | <i>Platanus racemosa</i>          | 2026-05-17 |
| white brodiaea                   | <i>Triteleia hyacinthina</i>      | 2026-05-15 |
| white-tipped clover              | <i>Trifolium variegatum</i>       | 2026-05-15 |
| Winecup Clarkia                  | <i>Clarkia purpurea</i>           | 2026-05-15 |
| Yellow Mariposa Lily             | <i>Calochortus luteus</i>         | 2026-06-11 |

**AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU**

Ecological Assessment Summary This site in Sonoma County supports a **mixed oak woodland and chaparral community**, dominated by blue oak and California black oak with understory contributions from California bay, California buckeye, and diverse herbaceous species. The presence of moisture-indicator species like Braun's Giant Horsetail and California wild rose alongside drought-adapted plants (dudleyas, cliff-brake ferns) suggests **variable soil moisture conditions with seasonal drainage patterns**, typical of foothill terrain with both seepage areas and well-drained slopes. For grading and drainage planning, expect localized zones of seasonal saturation during winter-spring runoff, requiring careful assessment of existing drainage patterns and consideration of erosion control measures to protect the oak woodland structure. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

**FIELD HAZARDS - Venomous & Toxic Species within 25mi**

|                 |                  |                  |              |
|-----------------|------------------|------------------|--------------|
| <b>636</b>      | <b>155</b>       | <b>91</b>        | <b>1439</b>  |
| Venomous Snakes | Venomous Spiders | Stinging Insects | Toxic Plants |

**VENOMOUS SNAKES**

| SPECIES                           | OBSERVATIONS |
|-----------------------------------|--------------|
| <i>Crotalus oreganus oreganus</i> | 636 obs      |

**VENOMOUS SPIDERS**

| SPECIES                     | OBSERVATIONS |
|-----------------------------|--------------|
| <i>Latrodectus hesperus</i> | 155 obs      |

**STINGING INSECTS**

| SPECIES                        | OBSERVATIONS |
|--------------------------------|--------------|
| <i>Dasymutilla aureola</i>     | 91 obs       |
| <i>Dasymutilla californica</i> | 91 obs       |
| <i>Dasymutilla sackenii</i>    | 91 obs       |

**TOXIC PLANTS**

| SPECIES                            | OBSERVATIONS |
|------------------------------------|--------------|
| <i>Toxicodendron diversilobum</i>  | 1100 obs     |
| <i>Datura wrightii</i>             | 57 obs       |
| <i>Datura stramonium</i>           | 57 obs       |
| <i>Datura ferox</i>                | 57 obs       |
| <i>Conium maculatum</i>            | 199 obs      |
| <i>Urtica gracilis</i>             | 83 obs       |
| <i>Urtica gracilis holosericea</i> | 83 obs       |

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Brief – Northern California Survey Site **Highest-Priority Hazards:** Poison oak (\*Toxicodendron diversilobum\*) is your dominant concern with 1,439 observations in the area —expect it everywhere, especially in brushy terrain. Northern Pacific rattlesnakes (\*Crotalus oreganus oreganus\*) are well-established here with 636 documented sightings, and black widow spiders (\*Latrodectus hesperus\*) present a secondary but real risk in sheltered spots. **Practical Avoidance:** Wear long pants, long sleeves, and closed boots; learn to identify poison oak's three-leaflet clusters and avoid contact entirely rather than relying on post-exposure washing. Stay alert for rattlesnakes in rocky outcrops and tall grass, especially during warm daylight hours—make noise while moving and watch hand placement on rocks. Check equipment, vehicles, and gear regularly for spiders and never work barefoot or with exposed skin in dense vegetation. **Seasonal Note:** Late spring through early fall brings peak snake activity and insect abundance; poison oak oils are potent year-round but cause worse reactions when the plant is actively growing (spring/summer). **Always verify current conditions locally before fieldwork.**

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

|            |              |           |
|------------|--------------|-----------|
| 3          | 545ft        | 3         |
| Beam Paths | Closest Beam | Fixed P2P |

| TX CALL | RX CALL | TYPE                 | PATH MI | CLOSEST |
|---------|---------|----------------------|---------|---------|
| WBO61   | WBO59   | Fixed Point-to-Point | —       | 545ft   |
| KYH37   | WQAT627 | Fixed Point-to-Point | —       | 927ft   |
| WRCR329 | WQAT627 | Fixed Point-to-Point | —       | 1289ft  |

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

|               |            |                |              |
|---------------|------------|----------------|--------------|
| 26            | 13         | 2              | 11           |
| Total Airways | Jet Routes | Victor Airways | Other Routes |

| AIRWAY | TYPE      | LEVEL | MEA EAST  | MEA WEST | TRACK  | REV TRACK |
|--------|-----------|-------|-----------|----------|--------|-----------|
| J1     | Jet Route | High  | 18,000 ft | —        | 303.8° | 121.4°    |
| J110   | Jet Route | High  | 18,000 ft | —        | 137.9° | 318.2°    |
| J126   | Jet Route | High  | 18,000 ft | —        | 325.5° | 144.0°    |
| J143   | Jet Route | High  | 18,000 ft | —        | 325.0° | 145.8°    |
| J3     | Jet Route | High  | 18,000 ft | —        | 342.8° | 161.8°    |
| J32    | Jet Route | High  | 18,000 ft | —        | 19.2°  | 199.6°    |
| J501   | Jet Route | High  | 18,000 ft | —        | 317.1° | 135.4°    |
| J58    | Jet Route | High  | 18,000 ft | —        | 65.3°  | 245.9°    |
| J65    | Jet Route | High  | 18,000 ft | —        | 304.1° | 121.1°    |
| J80    | Jet Route | High  | 18,000 ft | —        | 65.3°  | 245.9°    |
| J84    | Jet Route | High  | 18,000 ft | —        | 50.8°  | 231.0°    |
| J88    | Jet Route | High  | 18,000 ft | —        | 308.0° | 127.3°    |
| J94    | Jet Route | High  | 18,000 ft | —        | 65.3°  | 245.9°    |
| Q1     | Q-Route   | High  | —         | —        | 334.5° | 154.1°    |
| Q120   | Q-Route   | High  | —         | —        | 38.1°  | 219.8°    |
| Q122   | Q-Route   | High  | —         | —        | 37.8°  | 219.6°    |
| Q124   | Q-Route   | High  | —         | —        | 37.8°  | 219.6°    |
| Q138   | Q-Route   | High  | —         | —        | 47.7°  | 229.0°    |
| Q3     | Q-Route   | High  | —         | —        | 164.0° | 344.6°    |
| Q5     | Q-Route   | High  | —         | —        | 161.4° | 341.7°    |
| T257   | T-Route   | Low   | —         | —        | 321.1° | 140.5°    |
| T259   | T-Route   | Low   | —         | —        | 41.6°  | 221.8°    |
| T261   | T-Route   | Low   | —         | —        | 31.4°  | 211.7°    |
| T263   | T-Route   | Low   | —         | —        | 308.8° | 128.2°    |



| AIRWAY | TYPE          | LEVEL | MEA EAST | MEA WEST | TRACK  | REV TRACK |
|--------|---------------|-------|----------|----------|--------|-----------|
| V107   | Victor Airway | Low   | 7,000 ft | —        | 295.4° | 114.2°    |
| V108   | Victor Airway | Low   | 4,500 ft | —        | 117.7° | 296.8°    |

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

#### RAILROAD PROXIMITY — FRA North American Rail Network

8  
Rail Lines

| OWNER / OPERATOR | TYPE     | TRACKS | PASSENGER | STRACNET | SUBDIVISION |
|------------------|----------|--------|-----------|----------|-------------|
| CFNR             | Other    | 1      | No        | No       | —           |
| CFNR             | Yard     | 1      | No        | No       | —           |
| CFNR             | Mainline | 1      | No        | No       | VALLEJO     |
| SMRT             | Mainline | 1      | No        | No       | BRAZOS JCT. |
| NVRR             | Mainline | 1      | No        | No       | MARTINEZ    |
| USG              | Other    | 1      | No        | No       | —           |
| CFNR             | S        | 1      | No        | No       | —           |
| NVRR             | Yard     | 1      | No        | No       | —           |

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

#### WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

20089646136  
Total ObservationsTotal SpeciesBirdsMammalsReptilesAmphibians

##### BIRDS (64 species)

| COMMON NAME             | Scientific Name               | OBS |
|-------------------------|-------------------------------|-----|
| California Towhee       | <i>Melozone crissalis</i>     | 7   |
| California Quail        | <i>Callipepla californica</i> | 6   |
| Wild Turkey             | <i>Meleagris gallopavo</i>    | 6   |
| Snowy Egret             | <i>Egretta thula</i>          | 6   |
| Western Bluebird        | <i>Sialia mexicana</i>        | 5   |
| Ash-throated Flycatcher | <i>Myiarchus cinerascens</i>  | 4   |
| Great Egret             | <i>Ardea alba</i>             | 4   |
| Red-tailed Hawk         | <i>Buteo jamaicensis</i>      | 4   |
| Black Phoebe            | <i>Sayornis nigricans</i>     | 4   |
| Northern Flicker        | <i>Colaptes auratus</i>       | 3   |
| Cooper's Hawk           | <i>Astur cooperii</i>         | 3   |
| Western Screech-Owl     | <i>Megascops kennicottii</i>  | 3   |
| Turkey Vulture          | <i>Cathartes aura</i>         | 3   |
| Mallard                 | <i>Anas platyrhynchos</i>     | 3   |
| Bushtit                 | <i>Psaltriparus minimus</i>   | 3   |
| + 49 additional species |                               |     |

##### MAMMALS (6 species)

| COMMON NAME             | Scientific Name            | OBS |
|-------------------------|----------------------------|-----|
| Coyote                  | <i>Canis latrans</i>       | 3   |
| Black-tailed Jackrabbit | <i>Lepus californicus</i>  | 2   |
| Mule Deer               | <i>Odocoileus hemionus</i> | 2   |
| Domestic Cat            | <i>Felis catus</i>         | 1   |
| Eastern Fox Squirrel    | <i>Sciurus niger</i>       | 1   |
| Striped Skunk           | <i>Mephitis mephitis</i>   | 1   |

##### REPTILES (13 species)

| COMMON NAME                  | Scientific Name                      | OBS |
|------------------------------|--------------------------------------|-----|
| Western Fence Lizard         | <i>Sceloporus occidentalis</i>       | 11  |
| Pacific Gopher Snake         | <i>Pituophis catenifer catenifer</i> | 7   |
| Northern Pacific Rattlesnake | <i>Crotalus oreganus oreganus</i>    | 6   |
| California King Snake        | <i>Lampropeltis californiae</i>      | 3   |

| COMMON NAME                  | Scientific Name              | OBS |
|------------------------------|------------------------------|-----|
| Pond Slider                  | Trachemys scripta            | 2   |
| Western Skink                | Plestiodon skiltonianus      | 2   |
| Ring-necked Snake            | Diadophis punctatus          | 1   |
| Common Garter Snake          | Thamnophis sirtalis          | 1   |
| Western Yellow-bellied Racer | Coluber constrictor mormon   | 1   |
| Pacific Ringneck Snake       | Diadophis punctatus amabilis | 1   |
| Northwestern Pond Turtle     | Actinemys marmorata          | 1   |
| Common Sharp-tailed Snake    | Contia tenuis                | 1   |
| North American Racer         | Coluber constrictor          | 1   |

AMPHIBIANS (6 species)

| COMMON NAME                 | Scientific Name         | OBS |
|-----------------------------|-------------------------|-----|
| Pacific chorus frog         | Pseudacris regilla      | 4   |
| Western Toad                | Anaxyrus boreas         | 3   |
| American Bullfrog           | Lithobates catesbeianus | 3   |
| Foothill Yellow-legged Frog | Rana boylei             | 3   |
| Rough-skinned Newt          | Taricha granulosa       | 1   |
| California Giant Salamander | Dicamptodon ensatus     | 1   |

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment This site in Sonoma County, California represents a mixed coastal foothills ecosystem with mosaic habitat including grassland, oak woodland, and riparian or wetland corridors, as evidenced by the diverse breeding and migratory bird community and the presence of water-dependent species such as egrets, pond turtles, and multiple amphibians. The wildlife assemblage indicates proximity to seasonal or perennial water features and relatively intact native habitat connectivity, though the presence of invasive species (Eastern Fox Squirrel, American Bullfrog, Domestic Cat) and introduced reptiles (Pond Slider) suggests moderate human disturbance and habitat fragmentation. Of regulatory concern, Foothill Yellow-legged Frogs and California Giant Salamanders are California Species of Special Concern requiring sensitive riparian habitat management, while the Northern Pacific Rattlesnake and various native reptiles warrant documented avoidance during ground-disturbing activities. The overall community composition—dominated by generalist species but retaining sensitive amphibians and native reptile diversity—suggests this site maintains functional ecological value for wildlife movement and breeding, though cumulative land use pressures warrant careful assessment of any proposed development's impacts to riparian zones, grassland habitat, and wildlife corridors. --- **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳️ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 • AVERAGE SUN HOURS

| Period            | Peak Hrs/Day | Notes                        |
|-------------------|--------------|------------------------------|
| Annual average    | 6.6 hrs      | Overall solar resource       |
| Summer average    | 9 hrs        | June–August                  |
| Winter average    | 3.9 hrs      | December–February            |
| Worst month (Dec) | 3.2 hrs      | Size battery storage to this |

2 • IDEAL SOLAR ARRAY PITCH

Optimal tilt = 38.3° (latitude) → 9/12 pitch

| Pitch | Angle | % of Optimal | Notes            |
|-------|-------|--------------|------------------|
| 4/12  | 18.4° | 97%          | good             |
| 5/12  | 22.6° | 97.6%        | good             |
| 6/12  | 26.6° | 98.2%        | ↔ excellent      |
| 7/12  | 30.3° | 98.8%        | ↔ excellent      |
| 8/12  | 33.7° | 99.3%        | ↔ excellent      |
| 9/12  | 36.9° | 99.8%        | ↔ annual optimal |
| 10/12 | 39.8° | 99.8%        | ↔ excellent      |
| 12/12 | 45°   | 99%          | ↔ excellent      |

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 • SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

| Surface                    | Winter | Summer | Design Note                          |
|----------------------------|--------|--------|--------------------------------------|
| South glass (SHGC 0.4)     | 51     | 31     | Minimize south glass                 |
| South glass (SHGC 0.6)     | 76     | 46     | Clear/uncoated glass                 |
| North glass (any)          | 10     | 13     | Negligible — diffuse only            |
| East / West glass          | 27     | 104    | ⚠ Minimize — afternoon load          |
| Roof — dark (absorb 0.95)  | 236    | 301    | Highest load surface                 |
| Roof — light (absorb 0.35) | 87     | 111    | Cool roof — significant reduction    |
| Roof — metal (absorb 0.25) | 62     | 79     | Best roof option for heat            |
| South wall — dark          | 94     | 69     |                                      |
| South wall — light         | 31     | 23     | Light color = major savings          |
| Hardscape — asphalt        | 181    | 301    | ⚠ Heat island — avoid near structure |
| Hardscape — concrete       | 124    | 206    | Also reflects onto adjacent walls    |
| Hardscape — light pave     | 67     | 111    | Recommended within 50ft              |

Pool surface

23

38

Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.  
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

| Pitch | Dark Roof | Light Roof | Reduction vs Flat      |
|-------|-----------|------------|------------------------|
| 0/12  | 301 BTU   | 111 BTU    | baseline (worst)       |
| 4/12  | 241 BTU   | 89 BTU     | -20% dark / -20% light |
| 6/12  | 196 BTU   | 72 BTU     | -35% dark / -35% light |
| 8/12  | 166 BTU   | 61 BTU     | -45% dark / -45% light |
| 12/12 | 135 BTU   | 50 BTU     | -55% dark / -55% light |

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

| Assembly          | Lag     | Peak Interior Load | Note                                       |
|-------------------|---------|--------------------|--------------------------------------------|
| 2x4 frame (R-15)  | 0.6 hrs | ~15:36 PM          | R-value only — no mass benefit             |
| 2x6 frame (R-23)  | 0.9 hrs | ~15:54 PM          | R-value only — no mass benefit             |
| 2x8 frame (R-30)  | 1.2 hrs | ~16:12 PM          | R-value only — no mass benefit             |
| 2x10 frame (R-37) | 1.5 hrs | ~16:30 PM          | good                                       |
| 2x12 frame (R-44) | 1.8 hrs | ~16:48 PM          | good                                       |
| ICF 6" core       | 7 hrs   | ~22:00 PM          | ✓ shifts past sunset — natural vent window |
| 8" CMU block      | 6 hrs   | ~21:00 PM          | ✓ shifts past sunset — natural vent window |
| 12" concrete      | 8.5 hrs | ~23:30 PM          | ✓ shifts past sunset — natural vent window |

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

| Tree Ht | Shadow Reaches | Summer hrs/day | Equinox hrs/day | Winter hrs/day | WUI note     |
|---------|----------------|----------------|-----------------|----------------|--------------|
| 10ft    | All year       | 1.1            | 1               | 1.2            | OK           |
| 20ft    | All year       | 2.1            | 1.9             | 2.4            | OK           |
| 30ft    | All year       | 3.1            | 2.9             | 3.7            | Zone 2 check |
| 40ft    | All year       | 4              | 3.8             | 5.2            | △ verify     |
| 50ft    | All year       | 4.8            | 4.6             | 7.3            | △ verify     |
| 60ft    | All year       | 5.6            | 5.5             | 9.3            | ✗ Zone 1     |

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.  
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.  
△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 • ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

| Standard        | Unbalanced (1/150) | Balanced† (1/300) | Notes                                 |
|-----------------|--------------------|-------------------|---------------------------------------|
| Code minimum    | 8 sq in            | 4 sq in           | Moisture protection only              |
| Thermal comfort | 24 sq in           | 12 sq in          | Keeps attic within 20°F of outdoor    |
| WUI mesh 1/16"  | 60 sq in           | 30 sq in          | 2.5× compensation — mesh clogs        |
| WUI intumescent | 24 sq in           | 12 sq in          | Self-closing — no compensation needed |

**Pitch factor:** 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45  
**Color factor:** Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45  
**Scale:** multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor  
†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)  
**WUI conflict:** Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.  
△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🔍 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan You're in a sweet spot for solar—with about 6.6 peak sun hours daily on average, your panels will generate steady power year-round, though you'll see a noticeable jump in summer (up to 9 hours) and a dip in winter (down to 3.9 hours). Your roof should ideally tilt at about a 38-degree angle, which happens to match a 9-in-12 pitch if you're building or retrofitting—this angle lets panels drink in the sun most efficiently throughout the year. Here's something most homeowners miss: a dark roof absorbs a tremendous amount of summer heat—think of it like parking a black car in the sun versus a white one—so switching to a light-colored roof can cut your cooling load in half, saving you real money on air conditioning. That west-facing glass is your hidden enemy in summer afternoons; the sun pounds it with heat, so consider shading it with an awning, shade screen, or even a deciduous tree (which conveniently drops its leaves in winter, letting warming sun through when you need it). Thermal mass is simply the ability of your walls and floors to soak up heat during the day and release it slowly at night—thicker materials like ICF (insulated concrete forms) hold heat much longer than standard wood framing, keeping your home more stable. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.



Technical Brief — for architects, engineers, and surveyors

### Structural Design Drivers

The IECC Zone 3C (Marine) classification with zero frost-free days indicates a mild coastal climate with minimal thermal cycling stress on structure. Wind, snow, and seismic loads are not provided in available data and cannot be assumed; design team must obtain site-specific wind speed per ASCE 7 (likely reduced from inland values given marine exposure) and confirm that snow load is negligible for this zone. The absence of these parameters in the intelligence suggests either a data gap or confirmation of minimal loading—clarify with county building department before finalizing load cases. This coastal zone typically drives decisions around moisture management and corrosion protection over structural magnitude, but load analysis cannot proceed without explicit wind and seismic values.

### Geotechnical & Foundation Considerations

Coombs gravelly loam on 2 to 5 percent slopes presents a well-drained soil profile in Hydrologic Group C, favorable for conventional shallow foundations. Absence of frost depth data prevents specification of minimum footing embedment; Napa County standard practice or site-specific soil boring must establish this value before design. The gravelly texture and good drainage eliminate concerns over differential settlement from saturation, but bearing capacity and  $\phi$  angle require geotechnical investigation. Group C hydrology indicates moderate runoff potential—surface grading must direct water away from foundations to maintain the well-drained condition reported.

### Hazard Assessment

No Quaternary faults exist within the search radius, and no subsidence basin data was identified, removing those specific hazard drivers. Seismic design parameters (PGA, spectral accelerations) are not provided in this intelligence package and must be obtained from USGS hazard maps and California seismic codes; proximity to San Francisco Bay seismic zone warrants attention. Wildfire and flood hazard data are not included; confirm with Cal Fire and FEMA flood maps as part of pre-design risk scoping.

### Site Context & Mining Operations

Three active extractive operations exist within 5 miles: Basalt Rock Co (CLY, 2 mi), Mt George Pumice Deposit (PUM, 3.1 mi), and Silva Pumice Quarry (PUM, 5 mi). Proximity to active quarries introduces potential vibration, dust, and traffic impact on site operations. Additionally, 200 MRDS deposits within ~35 miles indicate mineral-rich geology; confirm whether the subject site itself overlies any recorded mineral claims or resource lands that could affect development rights or future use. No active oil or gas wells exist within 10 miles (nearest Mobil well at 7.2 mi is plugged), eliminating subsurface hydrocarbon hazard.

### Bottom Line

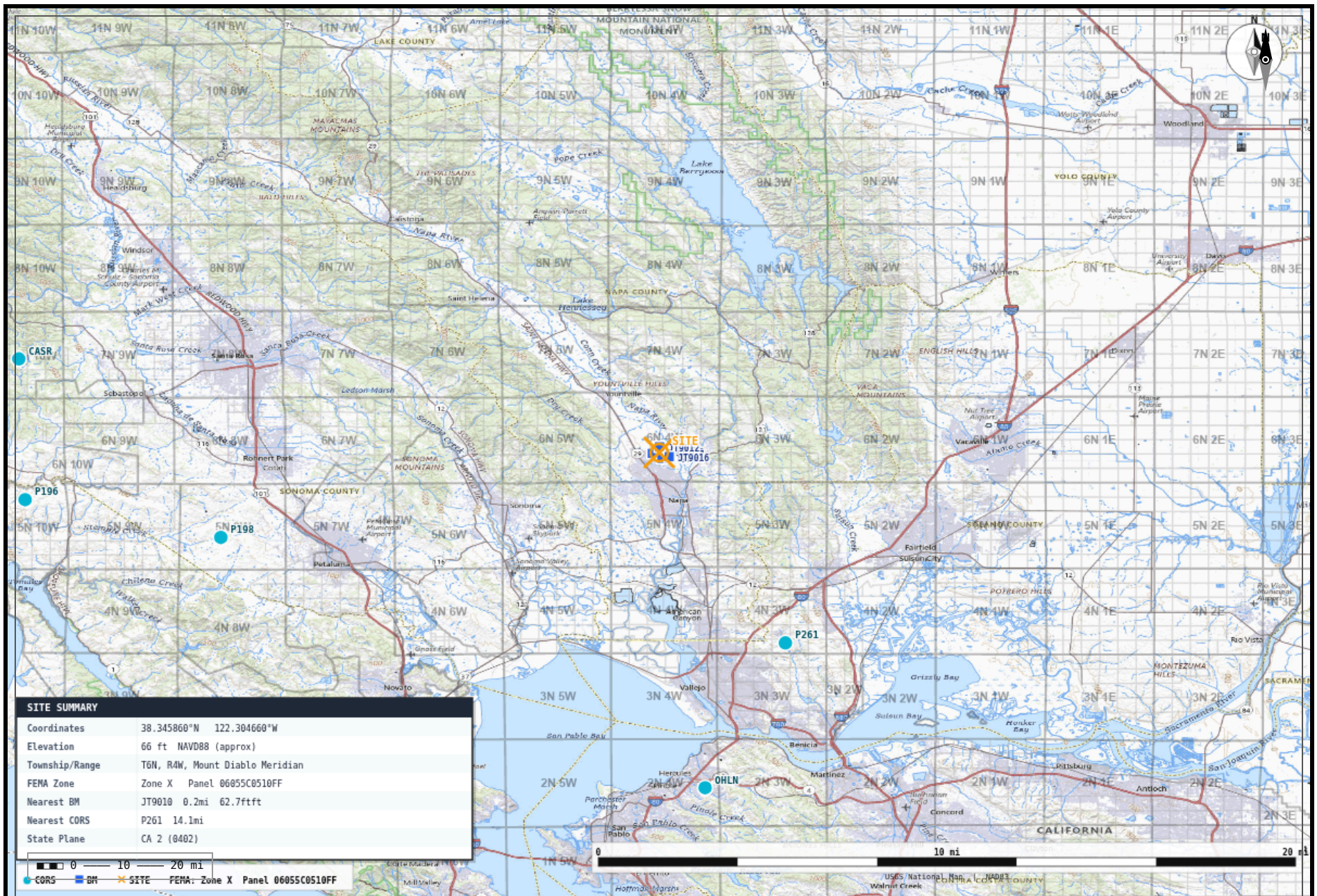
Obtain site-specific seismic design parameters, wind speed, frost depth, and geotechnical boring data before advancing structural system selection, and confirm mineral claims status and quarry activity impacts on site planning.



Copy to clipboard

Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.





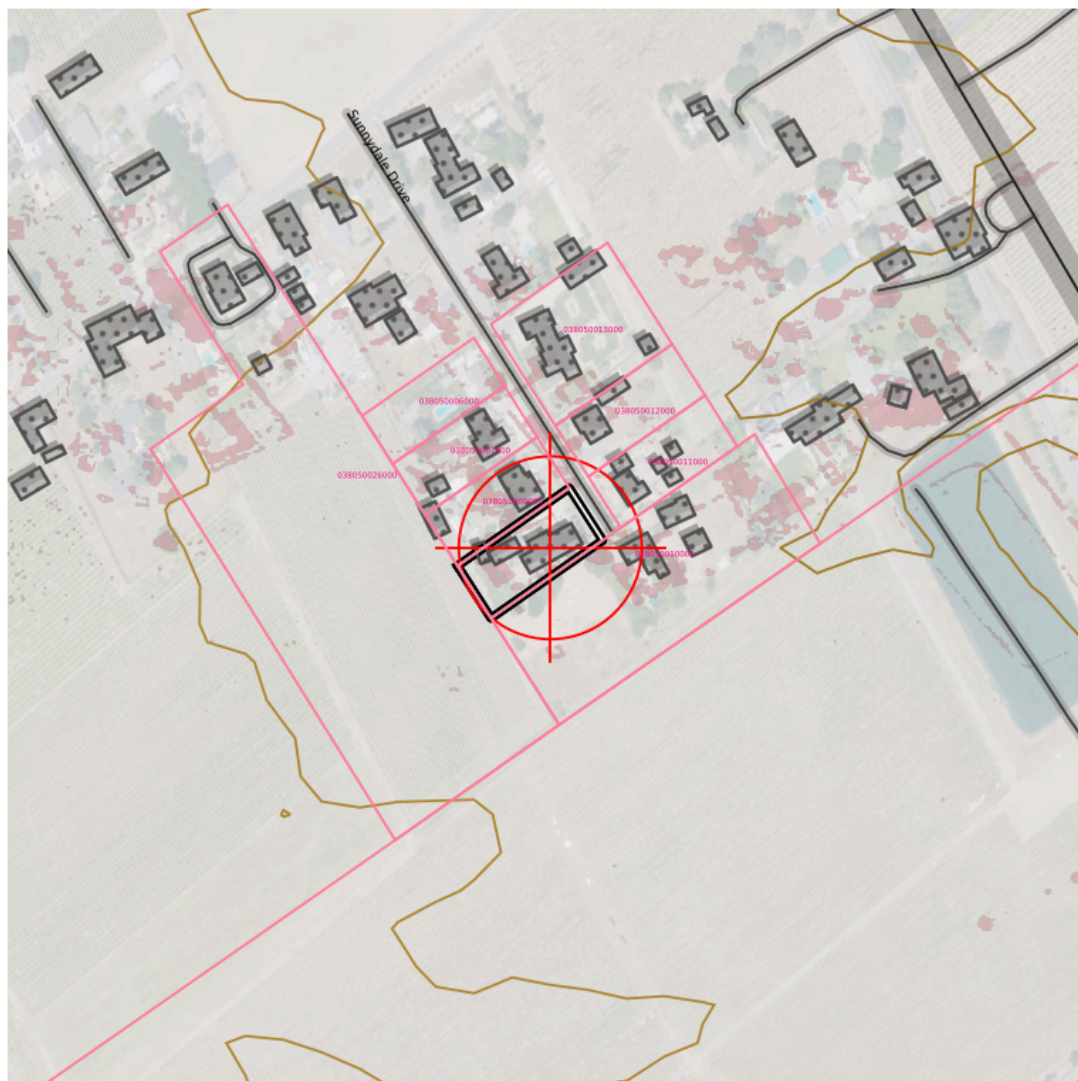
**TOPOSCOUT**  
FIELD MAP

**TopoScout Field Map**  
Lat: 38.34586 Lng: -122.30466  
38.345860°N 122.304660°W | Radius: 30 mi

Generated: July 14, 2026  
Base: USGS National Map  
Generated by TopoScout - outkastdesigns.com

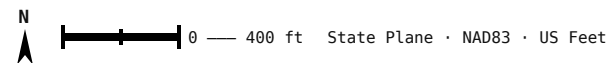


## TOPOSCOUT SITE PLAN PREVIEW



## INCLUDED LAYERS

■ County Boundary   ■ OTLS Grid   ■ Site Info   ■ Elevation Labels   ■ 3D Polylines (all geometry)





**SITE AERIAL – 600 ft × 600 ft**

38.34586°N, 122.30466°W · Google Aerial





**AREA CONTEXT — 3000 ft × 3000 ft**

38.34586°N, 122.30466°W · Google Aerial



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## ✂ FIELD RESOURCE LINKS

### PRE-MISSION PLANNING

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#### Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

#### FAA B4UFLY –

[https://www.faa.gov/uas/recreational\\_flyers/where\\_can\\_i\\_fly/b4ufly](https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly)

Drone airspace authorization, TFRs, restricted areas

#### Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

#### NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

### POST-PROCESSING

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#### NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

#### NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

#### CORS Data Download – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Raw RINEX observation files from CORS stations

### FEMA & FLOOD

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#### FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

#### FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

### BASE STATION & NETWORK

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#### NOAA CORS Network – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Continuously operating reference stations, data download

#### NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

#### IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

### REFERENCE & CALCULATIONS

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#### NGS Datasheets – <https://www.ngs.noaa.gov/cgi-bin/datasheet.prl>

Benchmarks, control points, passive monuments

#### NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

#### NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

#### NGS Geoid Models – [https://geodesy.noaa.gov/GEOID/](https://geodesy.noaa.gov/GEOID/GEOID18)

[GEOID18](https://geodesy.noaa.gov/GEOID/GEOID18) and current model downloads, GEOID calculator

### DRONE OPERATIONS

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#### FAA Part 107 — Commercial Rules –

[https://www.faa.gov/uas/commercial\\_operators](https://www.faa.gov/uas/commercial_operators)

Full Part 107 regulatory summary for commercial UAS operations

#### FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

#### FAA B4UFLY –

[https://www.faa.gov/uas/recreational\\_flyers/where\\_can\\_i\\_fly/b4ufly](https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly)

Pre-flight airspace check, TFRs, LAANC authorization

#### FAA Part 107 Waivers –

[https://www.faa.gov/uas/commercial\\_operators/part\\_107\\_waivers](https://www.faa.gov/uas/commercial_operators/part_107_waivers)

Waiver applications for operations beyond standard 107 rules

#### Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

### AIR QUALITY & WILDFIRE SMOKE

#### [AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

#### [AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

#### [NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

